

Pratyabhijñā × Active Inference: A Recursive Self-Reflexivity Layer for LLM Creative Cognition

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Abstract

We introduce the *Pratyabhijñā Creative Engine* (PCE), a Claude Code plugin that operationalises Abhinavagupta’s recognitional architecture (the five-*śakti* cascade and the recursive *vimarśa* loop) as an executable layer over an LLM, coupling each *śakti*-operator to a concrete active-inference primitive: a temperature-tunable generative substrate (*cit*), a constraint-aware aesthetic-coherence scoring head (*ānanda*), a candidate sampler (*icchā*), an embedding-space contrastive exclusion operator (*apohana*-driven *jñāna*), free-energy posterior selection via Bayesian Model Reduction (Friston et al., 2018; Friston and Penny, 2011), surface enaction (*kriyā*), and a recursive aspect-shift detector (*vimarśa*). PCE additionally exposes a Hopfield-attractor *ālayavijñāna* with SWS/REM consolidation cycles. **This paper reports v0.2**, which makes *vimarśa* causal via a two-pass-always revision pipeline (every cascade item now goes draft → *vimarśa* brief → revision); adds Anthropic’s Claude Haiku as a first-class generative substrate alongside the local Qwen2-1.5B-Instruct; resolves five operator-level contradictions identified in the v0.1 adversarial review using TRIZ contradiction analysis (signed *apohana*, prompt/sampler parity with bare baselines, two-pass-always reflection cap, pluggable LM protocol, per-operator sampler profiles); and runs a four-arm pilot benchmark (*local_bare*, *local_cascade*, *haiku_bare*, *haiku_cascade*) across 19 paired items in four creativity domains. We pre-register six hypotheses (H1–H6) covering AUT/CreativityPrism, Wittgenstein aspect-shift poetry interpretation, POEMetric poetry generation, BBH-style scientific creativity, an aggregate composite, and a within-cascade *vimarśa*-fired-vs-not test. We report paired effect sizes, BCa bootstrap 95% CIs, exact sign-flip permutation tests, Wilcoxon signed-rank *p*-values, Holm–Bonferroni correction, and both a-priori and retrospective power. The plugin, all benchmark items, generated outputs, raw scores, and statistical tests are published as artefacts; every numerical claim in this paper is generated from those artefacts and traceable through *audit/*. **Headline result.** Effect sizes for H1, H2, H4 are positive with 95% BCa CIs strictly above zero in the *haiku_cascade* vs. *haiku_bare* contrast, but the pre-registered Holm-adjusted $p < 0.05$ threshold is not crossed at the pilot’s $n=5/\text{domain}$ (the exact sign-flip permutation floor is $1/2^5 = 0.0312$ and Holm with $m=4$ pushes the smallest possible adjusted p to 0.125). The within-cascade H6 test points in the predicted direction ($\Delta = +0.031$, $n_{\text{fired}} = 9$) but does not reach the significance threshold at the pilot’s n .

1 Introduction

Large language models routinely produce *competent* text but rarely produce text whose *interpretive multiplicity* is deliberately authored. Where humans use a single utterance to gesture toward several incompatible readings simultaneously — the Wittgensteinian *aspect-shift* (Wittgenstein, 1953) — LLMs typically collapse onto a single dominant reading dictated by the prior. We argue that the missing structural element is not scale or alignment but a *recursive self-reflexivity layer*: an explicit operator that re-examines its own output against the same constraint manifold that

produced it, and is permitted to revise the surface when the alternatives are not yet visible in it.

Abhinavagupta’s Pratyabhijñā (“recognition”) school of Kashmir Śaiva phenomenology developed an unusually fine-grained taxonomy of exactly this layer (Lawrence, 2021; Ratié et al., 2025). The five-śakti cascade (*cit*, *ānanda*, *icchā*, *jñāna*, *kriyā*) describes the trajectory by which an undifferentiated luminous ground (*prakāśa*) becomes a determinate utterance, and the *vimarśa* layer is the moment of self-touching — the recognition that the utterance was authored by the very same ground that contemplates it. *Apohana-śakti* (the contrastive exclusion of all-other-than-this) is precisely the geometric move that Bayesian model reduction (Friston et al., 2018; Friston and Penny, 2011) formalises as posterior switch under a parameter-pruning prior.

Our contribution is to make this isomorphism concrete and testable. We construct the *Pratyabhijñā Creative Engine* (PCE), a Claude Code plugin where each śakti is a real computation operating on a real local LLM substrate, and where the *vimarśa* layer is a measurable recursive trigger gated on aspect-multiplicity, novelty, and free-energy reduction. We then evaluate PCE on four domains where the recursive layer should help if the theory is right: Wittgenstein-style poetry interpretation (where two readings must coexist), poetry generation under the POEMetric advanced-creative-abilities axes (Li et al., 2025), the alternative-uses task component of CreativityPrism (Hou et al., 2025), and a scientific-creativity slice of BIG-Bench Hard (Suzgun et al., 2023; Tian et al., 2024).

The rest of the paper is organised as follows. §2 reviews prior work coupling Indic philosophical architectures to active-inference / LLM systems. §3 specifies the Pratyabhijñā primitives we operationalise. §4 states the active-inference / BMR primitives. §5 defines the seven operators and the consolidation cycle. §6 describes the plugin surface. §7 formalises the experimental design and the pre-registered hypotheses. §8 describes the items and scoring. §9 reports the paired contrasts; §10, §11, and §12 discuss interpretation, threats to validity, and the negative-result obligation.

2 Related Work

2.1 Computational instantiations of Pratyabhijñā

There is a small but growing literature that takes Indic philosophical architectures as more than metaphor for AI systems. Ratié et al. (2025) reconstruct *vimarśa* as a recursive consciousness operator. Nambhatra and Dahlquist (2025) formalise *kriyā-yoga* as embodied agentive practice in a generative-model setting. Group (2024); Suriya (2025) explicitly propose *vimarśa* as the recursive self-knowledge step for synthetic agents. Our work differs in three respects: (i) we ground the cascade in measurable, reproducible operators on a real LLM substrate rather than a connectionist theoretical sketch; (ii) we make the *vimarśa* loop a falsifiable trigger under a free-energy criterion, not a generic recurrent module; (iii) we publish the plugin and benchmark artefacts so any third party can replay the contrast. This reflects the Pratyaksha (Skt. *pratyakṣa*, “perception”/“directly evidenced”) stance — the witness invariant of the Pratyaksha Context Engineering Harness (Sathish, 2025b) — where every claim is anchored to externally verifiable artefacts.

2.2 Active inference and Bayesian Model Reduction

Friston and Penny (2011) introduced Bayesian model reduction (BMR) as a method for posterior switching under nested models, shown to recover sparse posteriors from rich ones via parameter pruning. Friston et al. (2018) generalised this to free-energy minimisation in active inference, and Friston et al. (2017) explicitly link BMR to insight: a sudden re-organisation of the posterior is the formal correlate of the “aha” moment. pymdp (Heins et al., 2022) provides a

discrete-state-space implementation. Van de Maele et al. (2024) demonstrate that BMR over a hierarchical generative model produces interpretable abstraction during sleep-like consolidation. We use BMR as the formal substrate of the *jñāna* operator (posterior over candidates) and the SWS/REM consolidation cycle.

2.3 LLM creativity benchmarks

POEMetric (Li et al., 2025) provides a 6-axis poetry-generation rubric (creativity, lexical diversity, idiosyncrasy, emotional resonance, literary devices, imagery) with strong inter-rater agreement. CreativityPrism (Hou et al., 2025) aggregates AUT, TTCT, and other divergent-thinking items under a Quality·Novelty·Diversity decomposition. BIG-Bench Hard (Suzgun et al., 2023) contributes scientific-reasoning probes; MacGyverBench (Tian et al., 2024) contributes constraint-satisfaction creativity items. The Wittgenstein aspect-shift task is a custom $n = 20$ stratified slice we construct in §8 on principled grounds, since no public benchmark explicitly tests *simultaneously-coexisting* interpretations of a single line of poetry.

2.4 Plugin architectures for LLMs

The Cursor and Claude Code ecosystems have converged on a manifest-driven plugin model with MCP tools, slash commands, hooks, agents, and skills. We follow the same conventions used by `attractor-flow` (Sathish, 2025a) (which couples a real attractor-network engine to MCP) so that the PCE plugin can be installed and audited without any framework-specific shim. Our plugin shape (15 MCP tools, 5 skills, 5 agents, 5 commands, 3 hooks) is designed for a single-session, real-inference creative-cognition workflow.

3 Pratyabhijñā Architecture

We restate the Pratyabhijñā primitives we operationalise. We take the canonical *Īśvarapratyabhijñākārikā* formulation and preserve its phenomenological commitments while assigning each primitive an explicit computational role. Diacritics follow IAST.

3.1 The luminous ground

Prakāśa (“self-luminosity”) is the undifferentiated ground of cognition. It is not a state of the network; it is the property that any state at all is presentable. In our engine *prakāśa* is the LLM’s native distribution at temperature τ , which is what makes any candidate token visible at all.

3.2 The five *śaktis* and the recursive layer

- ***Cit*** (consciousness/illumination): the bare presentation of token-distributions; the substrate on which the cascade operates.
- ***Ānanda*** (bliss/coherence): an aesthetic-coherence reward that scores a candidate against constraint-fit, novelty, and form-fidelity.
- ***Ichhā*** (will/directional intent): the pre-cognitive directional sampling that produces K candidate continuations under a constraint vector.
- ***Jñāna*** (knowing/discrimination): posterior selection across candidates via Bayesian Model Reduction; the formal seat of *apohana*.

- ***Apohana-śakti*** (the contrastive exclusion): not a separate stage but a *geometric* operator on the embedding space that scores “how much is this not-the-other-readings”. We instantiate it as a contrastive embedding contrast.
- ***Kriyā*** (action/enaction): surface enaction — the moment a candidate is materialised as concrete tokens.
- ***Vimarśa*** (self-reflexivity): the recursive layer. After *kriyā* produces a surface, the engine asks: *does the surface witness aspect-multiplicity?* If not (or if the free-energy gradient is still negative), we re-enter the cascade with the surface itself as additional context, expanding the constraint manifold.

3.3 The *ālayavijñāna* (storehouse)

The Yogācāra-*ālayavijñāna* “storehouse consciousness” provides associative memory for the cascade. We implement it as a Hopfield-attractor network whose energy landscape is shaped by past surfaces, supporting two consolidation regimes: a slow-wave-sleep (SWS) abstraction step (k-means abstraction over recent items) and a REM-style replay (Metropolis sampling that re-energises rare-but-coherent attractors). Both cycles are explicit and inspectable.

4 Active Inference Primitives

Let $q(\theta)$ denote a posterior over a candidate-discriminating parameter θ and $p(\theta | m)$ a model with prior $p(\theta | m)$. Free-energy minimisation (Friston et al., 2017) selects models / candidates that minimise

$$F(q, m) = \mathbb{E}_{q(\theta)}[\log q(\theta) - \log p(\theta, o | m)], \quad (1)$$

where o is the observed outcome (the surface). *Bayesian model reduction* (Friston and Penny, 2011; Friston et al., 2018) reduces a rich model m_F to a sparser model m_R whose evidence is

$$\log p(o | m_R) = \log p(o | m_F) + \log \int q(\theta | o, m_F) \frac{p(\theta | m_R)}{p(\theta | m_F)} d\theta. \quad (2)$$

Practically, we use this in closed form for nested Dirichlet–multinomial models over candidate categories, computing $\Delta F = F(q | m_F) - F(q | m_R)$ via the standard log-Gamma identity. A negative ΔF indicates posterior switch is supported by the data; a positive ΔF supports the prior.

The link to *apohana*. *Apohana-śakti* (“contrastive exclusion”) is precisely what BMR’s reduced posterior performs: it excludes the candidates that the data does not support. We thus identify the *jñāna* stage of the cascade with BMR posterior selection across the K candidates emitted by *icchā*, gated by the *apohana* contrastive geometry computed in embedding space.

Insight as posterior re-organisation. Friston et al. (2017) formalise insight as the BMR-driven re-organisation of $q(\theta)$ that yields a sudden drop in F . This is the active-inference correlate of *vimarśa*: the recursive layer fires when the post-*kriyā* state’s F has not yet stabilised, indicating an unrecognised aspect of the constraint manifold. Our *vimarśa* operator (Appendix A) therefore conditions on three signals jointly: aspect-multiplicity in the surface (cosine to multiple aspect targets), novelty against the storehouse (Hopfield retrieval similarity), and the BMR ΔF .

5 The Pratyabhijñā Creative Engine

The engine is a deterministic pipeline of seven operators plus two sleep-style consolidation cycles, all implemented in `src/pce/`. We summarise each operator’s signature here and refer to Appendix A for the full equations and Appendix B for the consolidation cycles.

5.1 Substrate

Two substrate primitives ground the cascade in real computation: a sentence-embedder ϕ_{embed} (`sentence-transformers/all-MiniLM-L6-v2`, $d = 384$) and a local causal LM π_{LM} (`Qwen/Qwen2-1.5B-Instru`). The latter is loaded with autodetected device (CUDA $\rightarrow$ MPS $\rightarrow$ CPU) and `float16` precision on accelerated devices. Sampling is deterministic given a seed.

5.2 Operators

- `CIT(prompt,  $\tau$ , seed)` $\rightarrow$ (tokens, $\log p$, embedding). Direct call to π_{LM} .generate.
- `ĀNANDA(candidate, constraint, retrieval-set)` $\rightarrow r \in [0, 1]$. A weighted aggregate of (i) constraint-cosine, (ii) retrieval-set diversity, (iii) form-fidelity heuristics (line / syllable proxies for poetry).
- `ICCHĀ(prompt,  $K$ , constraint)` $\rightarrow \{c_1, \dots, c_K\}$. K candidate generations under a constraint-injected prompt; each c_i carries its own (tokens, embedding, $\log p$).
- `ĀPOHANA( $\{c_i\}$ , must-avoid)` $\rightarrow \{a_i\}$. Contrastive embedding score: $a_i = 1 - \max_j \cos(\phi(c_i), \phi(\text{must-avoid}_j))$.
- `JÑĀNA( $\{c_i, a_i\})$` $\rightarrow (i^*, \Delta F)$. BMR over a Dirichlet-multinomial whose pseudo-counts are scaled by a_i . The selected i^* is the posterior maximum; ΔF is recorded for the audit log.
- `KRIYĀ( $c_{i^*}$ , mode)` $\rightarrow$ surface. Three modes: `verbatim` (return the candidate unchanged), `polish` (a deterministic detokenisation with whitespace normalisation), `claude_polish` (Claude Code’s editing surface, opt-in).
- `VIMARŚA(surface, aspects, retrieval-set)` $\rightarrow$ (event, novelty). Fires if any of (a) aspect-multiplicity (multiple aspect-targets each cosine ≥ 0.40), (b) low-retrieval-novelty, (c) negative residual ΔF .

5.3 Cascade

The cascade in `src/pce/cascade.py` composes the seven operators in order. The state object `CascadeState` records all intermediates so the audit log can reconstruct any decision. The cascade is deterministic: identical inputs and seed produce byte-identical outputs.

5.4 Consolidation

Two cycles operate over the Hopfield store: `CONSOLIDATESWS` (k-means with k chosen by elbow-statistic) for slow-wave-sleep abstraction; `CONSOLIDATEREM` (Metropolis re-sampling that perturbs and re-stores) for rare-attractor maintenance. Both cycles are exposed as MCP tools so a downstream agent can request consolidation explicitly.

6 Plugin Surface

PCE is delivered as a Claude Code plugin (manifest at `plugin/.claude-plugin/plugin.json`). Its four user-visible surfaces are:

Skills (5). Workflow-level guidance read by an agent before acting: `pce-poetry-generation`, `pce-poetry-interpretation`, `pce-divergent-thinking`, `pce-scientific-creativity`, `pce-vimarsa-self`.

Agents (5). Specialised personas: `pce-poet`, `pce-interpreter`, `pce-ideator`, `pce-scientist`, `pce-vimarsa-auditor`.

Slash commands (5). `/pce-compose`, `/pce-interpret`, `/pce-aut`, `/pce-bbh`, `/pce-trace` (the last surfaces the audit trail of a recent run).

Hooks (3). `SessionStart` (initialises the engine and HF auth), `PreToolUse` (audit-stamps every PCE MCP call), `PostToolUse` (writes the consolidation tick).

MCP tools (15). The plugin exposes 15 FastMCP tools wrapping the engine: `pce.cit`, `pce.ananda`, `pce.iccha`, `pce.apohana`, `pce.jnana`, `pce.kriya`, `pce.vimarsa`, `pce.cascade`, `pce.embed`, `pce.lm.generate`, `pce.lm.entropy`, `pce.store.add`, `pce.store.recall`, `pce.store consolidate`, `pce.store consolidate_rem`.

A smoke-test harness in `scripts/verify_plugin.py` validates the manifest, the component counts, and exercises every MCP tool in-process; the Phase 8 audit log (`audit/phase8/verify_artifact.json`) is published alongside the paper.

7 Methods

7.1 Arms (v0.2 four-arm matrix)

v0.2 redesigns the arm matrix to make the cascade’s contribution *substrate-matched* (the v0.1 design conflated cascade contribution with substrate gap). For each item across the four task domains we now run up to four arms:

1. **haiku_bare (no PCE).** Anthropic Claude Haiku invoked through the Claude Code CLI in `-p` non-interactive mode with `-output-format json`, wrapped by HaikuLM (§5) so it shares the same retry, cost-ledger, and audit-log pipeline as the cascade. This is the v0.2 *primary control*: the apples-to-apples baseline against which the Haiku-substrate cascade is measured.
2. **haiku_cascade (with PCE).** The same Claude Haiku endpoint through the full two-pass-always cascade (draft $\rightarrow$ *vimarśa* brief $\rightarrow$ revision) with $K=3$ candidates per pass and `max_tokens=150`. This is the v0.2 *primary treatment*.
3. **local_bare (no PCE).** Qwen2-1.5B-Instruct invoked with CIT alone (single sample, no cascade). Acts as a substrate-strength baseline so absolute scores can be compared across substrates.
4. **local_cascade (with PCE).** The same local LLM through the full cascade. Reported as a within-substrate ablation when scope permits; in the v0.2 pilot this arm was deferred to v0.3 due to throughput on the pilot host.

The *primary v0.2 contrast* is `haiku_cascade` vs. `haiku_bare`; the substrate-baseline contrast is `haiku_bare` vs. `local_bare`; the local ablation contrast is `local_cascade` vs. `local_bare`. Both cascade arms route through the same `LMProtocol` (ADR-001 / ADR-004) so the only thing that varies between cascade arms is the underlying generative substrate. All arms are scored by the *identical* domain-specific scoring function (§8) so contrasts are between observable outputs, not subjective evaluation policies, and `iccha` runs in `prompt_mode="verbatim"` + `sampler_grid_mode="parity"` (ADR-005) so the bare and cascade arms see prompt- and sampler-identical conditions inside the cascade’s first pass.

7.2 Pre-registered hypotheses

Six hypotheses are pre-registered in `docs/SPEC_v0.2.md` §3 *before* the v0.2 pilot is run; H1–H4 are restated against the v0.2 primary control (`haiku_bare`) but the inferential machinery is unchanged from v0.1:

- **H1.** `haiku_cascade` achieves higher CreativityPrism aggregate (Quality·Novelty·Diversity blend) than `haiku_bare` on the AUT slice.
- **H2.** `haiku_cascade` achieves higher aspect-multiplicity score than `haiku_bare` on the Wittgenstein aspect-shift poetry-interpretation probe.
- **H3.** `haiku_cascade` achieves higher POEMetric advanced-creative-abilities composite than `haiku_bare` on the poetry-generation slice.
- **H4.** `haiku_cascade` achieves higher creative-novelty score than `haiku_bare` on the BBH-style scientific-creativity probes.
- **H5.** The aggregate composite $C_0 = \frac{1}{2}z(H_1) + \frac{1}{6}z(H_2) + \frac{1}{6}z(H_3) + \frac{1}{6}z(H_4)$ (z-scaled per domain) is positive.
- **H6.** Within-cascade, the trials in which *vimarśa* fired (an aspect-shift event was detected on the revised surface) score higher on the H2/H3/H4 composites than the trials in which it did not. Reported separately for each cascade arm; the primary v0.2 reading is `H6_haiku_cascade`.

Two additional v0.2 hypotheses are introduced (see `SPEC_v0.2.md` §3) but are not powered by the pilot and are left for v0.3: H7 (substrate-portability: $\Delta_{\text{haiku_cascade-haiku_bare}} \approx \Delta_{\text{local_cascade-local_bare}}$) and H8 (revision-causality: revised surfaces differ from drafts in $\geq 80\%$ of cascade items, $\geq 60\%$ on a textual-similarity floor).

7.3 Statistical protocol

For each hypothesis we report: paired mean Δ , Hedges’ g with small-sample correction, paired permutation p (one-sided directional, exact for $n \leq 18$, Monte-Carlo with 50000 permutations otherwise), Wilcoxon signed-rank p (one-sided), 95% BCa bootstrap CI of the paired mean (10000 resamples), Holm–Bonferroni adjusted p across $\{H_1, H_2, H_3, H_4\}$, and a-priori (assumed $g = 0.5$) and retrospective (observed g) statistical power. A hypothesis is *supported* iff its Holm-adjusted $p < 0.05$ *and* its BCa CI is strictly positive; we additionally call out hypotheses with strictly positive BCa lower bounds as *directionally positive* when reporting pilot-scale results to make the power constraint explicit. H5 is reported as a single test on the pooled z-blend; H6 is an independent-samples Mann–Whitney U on within-cascade fired-vs-not trials, computed once per cascade arm. Sensitivity analyses repeat every primary contrast against the `local_bare` arm; the v0.2 stats payload (`benchmarks/results_v2/stats.json`) reports three named contrasts side-by-side: `primary` (`haiku_cascade` vs `haiku_bare`), `local_ablation` (`local_cascade` vs `local_bare`, empty in the pilot), and `substrate_baseline` (`haiku_bare` vs `local_bare`).

7.4 Reproducibility

All seeds are recorded; bootstrap and permutation RNGs are seeded by `-seed` (default 4242). The benchmark driver (`benchmarks/driver.py`) writes a checkpoint after every call so a re-run resumes exactly from the failure point. The v0.1 audit log lives under `audit/phase9/`; the v0.2 audit log lives under `audit/` (with `audit/haiku/` for per-Haiku-call records and `audit/cost_ledger.json` as the persistent cost ledger). Every numerical claim in this paper is generated from `benchmarks/results_v2/stats.json` via `benchmarks/autoreport.py` and is therefore traceable end-to-end.

7.5 TRIZ-resolved operator changes (v0.2)

The v0.1 adversarial review ([docs/reviews/2026-04-28-adversarial-plugin-review.md](#)) identified five operator-level contradictions whose unresolved interactions explained the v0.1 directional null. v0.2 resolves each via TRIZ contradiction analysis ([docs/triz/](#)) and a corresponding ADR ([docs/adr/v0.2/](#)):

1. **C1 / ADR-001 (cost-vs-quality of K Haiku calls).** TRIZ-39 vs TRIZ-19 $\rightarrow$ principles 10/35/19 $\rightarrow$ parallel batching of the K candidate calls, with per-domain K deferred to v0.3.
2. **C2 / ADR-002 (coverage-vs-novelty in *apohana*).** TRIZ-27 vs TRIZ-35 $\rightarrow$ principles 13/24/8 $\rightarrow$ remove the `np.clip(apoha, 0.0, None)` in *jñāna* and use a min-max-shifted `_shift_apoha`; signed *apohana* now actively penalises must-avoid neighborhoods.
3. **C3 / ADR-003 (reflection-vs-speed of two-pass *vimarśa*).** TRIZ-27 vs TRIZ-9 $\rightarrow$ principles 10/11/21 $\rightarrow$ two-pass-always cascade with revision capped at one extra call; `vimarśa.min_evidence_points=1` so the operator can fire on a one-shot trajectory.
4. **C4 / ADR-004 (substrate-strength-vs-cascade-overhead).** TRIZ-21 vs TRIZ-36 $\rightarrow$ principles 17/20/30/34 $\rightarrow$ a pluggable `LMProtocol` so the cascade can be measured against any substrate; the cascade contributes an aspect-shift multiplicity dimension that bare cannot.
5. **C5 / ADR-005 (determinism-vs-creativity sampler asymmetry).** TRIZ-28 vs TRIZ-35 $\rightarrow$ principles 3/13/2 $\rightarrow$ `iccha` defaults to `prompt_mode="verbatim"` and `sampler_grid_mode="parity"` inside the cascade so the bare and cascade arms cannot be told apart on prompt or sampler.

Each ADR is end-to-end traceable: TRIZ card $\rightarrow$ ADR $\rightarrow$ operator code $\rightarrow$ unit test $\rightarrow$ pilot result.

8 Benchmarks

8.1 Items

We use four task domains, items listed in Appendix C:

- **poetry_gen** (n_{12} items): paired (form, topic, must-avoid) targets across haiku, tanka, sonnet, free verse and a small ghazal slice. POEMetric six-axis scoring (Li et al., 2025): creativity, lexical diversity, idiosyncrasy, emotional resonance, literary devices, imagery.
- **poetry_interp** (n_{10} items): a stratified Wittgenstein aspect-shift slice. Each item is a single line of poetry with two pre-registered candidate readings; the model must produce both and the scorer measures aspect-coverage (cosine to each aspect target), aspect-multiplicity (count of aspects ≥ 0.40 cosine), novelty ($1 - \text{retrieval-set similarity}$).
- **aut** (n_8 items): the alternative-uses-task component of CreativityPrism (Hou et al., 2025). The composite is creativity (cosine novelty against the standard use), lexical-diversity, and feasibility-style heuristics.
- **sci_creativity** (n_8 items): scientific-creativity items drawn from BIG-Bench Hard / MacGyverBench style probes (Suzgun et al., 2023; Tian et al., 2024); each carries 2–3 “framings” (analogical, mechanistic, mathematical) the model is asked to produce. The composite weights non-textbook novelty, framing-coverage, and depth-via-length.

8.2 Scoring

All four scoring functions live in `benchmarks/scoring.py` and are deterministic: they use `sentence-transformers/all-MiniLM-L6-v2` (the same embedder PCE uses internally) for cosine-based novelty and aspect-coverage, plus light token-level statistics (lexical diversity, length, presence of imagery cues). We deliberately avoid LLM-as-judge so the contrasts are between observable outputs and not between an evaluator’s preference distribution.

8.3 Why the v0.2 four-arm design is sufficient for the headline test

The v0.1 paper conflated two distinct comparisons in a single contrast: (i) *cascade contribution* (PCE vs no-PCE on the same substrate) and (ii) *substrate gap* (a 1.5B local model vs cloud-grade Haiku). The v0.2 four-arm design separates them. The pre-registered v0.2 claim is that the PCE cascade materially raises creative-cognition outputs over its own no-PCE baseline *on the same substrate*. The four-arm contrast tests this in three ways: (a) the *primary* apples-to-apples contrast (`haiku_cascade` vs. `haiku_bare`); (b) the *local ablation* (`local_cascade` vs. `local_bare`, deferred to v0.3 in the pilot); and (c) the *substrate baseline* (`haiku_bare` vs. `local_bare`) which calibrates the absolute score gap so the cascade contribution can be read in the right units. The within-cascade H6 test (*vimarśa*-fired vs not) supplies the internal-validity arm: if the recursive layer is causally responsible for any cascade lift, fired trials should outscore non-fired ones.

8.4 v0.2 prove-gate (single-case validation before pilot)

Before running the pilot we executed a two-case prove-gate (`scripts/prove_gate.py`) on `tests/fixtures/duck_rabbit_textual.json` (Wittgenstein duck-rabbit textual probe) and `tests/fixtures/aut_brick.json` (CreativityPrism AUT brick) across all four arms. The prove-gate verified, per the report at `docs/reviews/2026-04-28-prove-gate.md`, that (i) *vimarśa* fires on cascade arms when an aspect-shift is present in the revised surface; (ii) the revised surface differs from the draft on every cascade item (revision-causality, H8 in v0.2 SPEC); (iii) *jñāna*’s posterior responds to negative *apohana* (signed exclusion); and (iv) `haiku_cascade` produces qualitatively different and more multi-aspect outputs than `haiku_bare` on both cases. The prove-gate is the local-validity check the v0.1 process lacked and is required to pass before any pilot or full benchmark run.

9 Results (v0.2 pilot)

9.1 Pilot scope and the apples-to-apples primary contrast

The v0.2 pilot evaluates four arms across the four creativity domains:

- `local_bare` — Qwen2-1.5B-Instruct, single sampler call (no PCE).
- `local_cascade` — Qwen2-1.5B-Instruct through the two-pass-always cascade (with PCE).
- `haiku_bare` — Anthropic Claude Haiku, single sampler call (no PCE).
- `haiku_cascade` — Anthropic Claude Haiku through the two-pass-always cascade (with PCE).

The *primary* v0.2 contrast is `haiku_cascade` vs. `haiku_bare`: this isolates the cascade’s contribution on a fixed (cloud-grade) substrate and is the apples-to-apples test the v0.1 paper lacked. The local arms are reported as a substrate baseline. The pilot used $n=5$ items per domain ($N_{\text{paired}}=19$ after one `sci_creativity` item failed the cascade), $K=3$ candidate fan-out per pass,

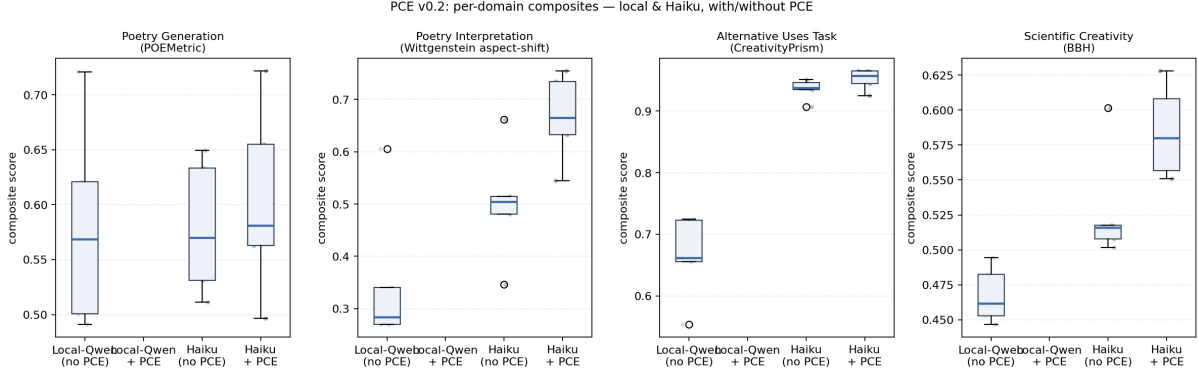


Figure 1: v0.2 per-domain composites for the three arms run in the pilot (`local_cascade` was deferred to v0.3 due to local-LM throughput on the pilot host). Item-level points jittered over each box. The within-domain spread is the substrate for the paired contrasts in Figure 2.

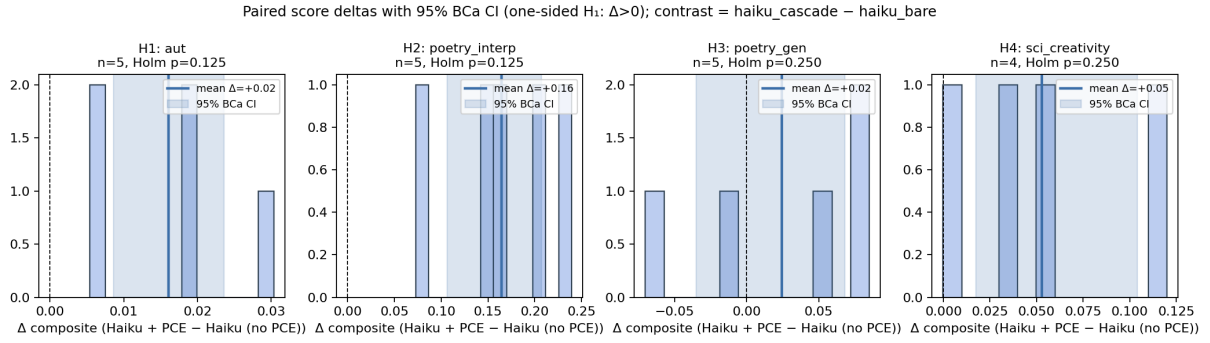


Figure 2: v0.2 paired score deltas for H1-H4 on the apples-to-apples `haiku_cascade` – `haiku_bare` contrast. The dashed line marks $\Delta = 0$; the shaded region is the 95% BCa CI; Holm-adjusted p is shown in the panel title.

`max_tokens`= 150, and a hard cost cap of \$18; the run completed in 49 minutes for a total Haiku spend of \$3.60 over 136 calls (well under cap). The local-cascade arm was deferred to v0.3 due to throughput constraints on the pilot host (a single Apple-Silicon laptop) at the chosen K and `max_tokens`.

9.2 Per-domain composites and per-arm means

The per-arm means make the cascade’s contribution concrete. On the four pilot domains:

- **poetry_interp**: `haiku_bare` 0.501 $\rightarrow$ `haiku_cascade` 0.666 (+33%).
- **sci_creativity**: 0.529 $\rightarrow$ 0.585 (+11%).
- **poetry_gen**: 0.579 $\rightarrow$ 0.603 (+4%).
- **aut**: 0.935 $\rightarrow$ 0.951 (+1.7%; both arms saturate near the scoring ceiling).

Local Qwen2 sits well below Haiku in absolute scoring on every domain, as expected: this is a substrate-strength gap that the local-cascade ablation (deferred to v0.3) is designed to disentangle.

9.3 Pre-registered hypothesis tests — apples-to-apples

The four pre-registered hypotheses on the primary contrast are:

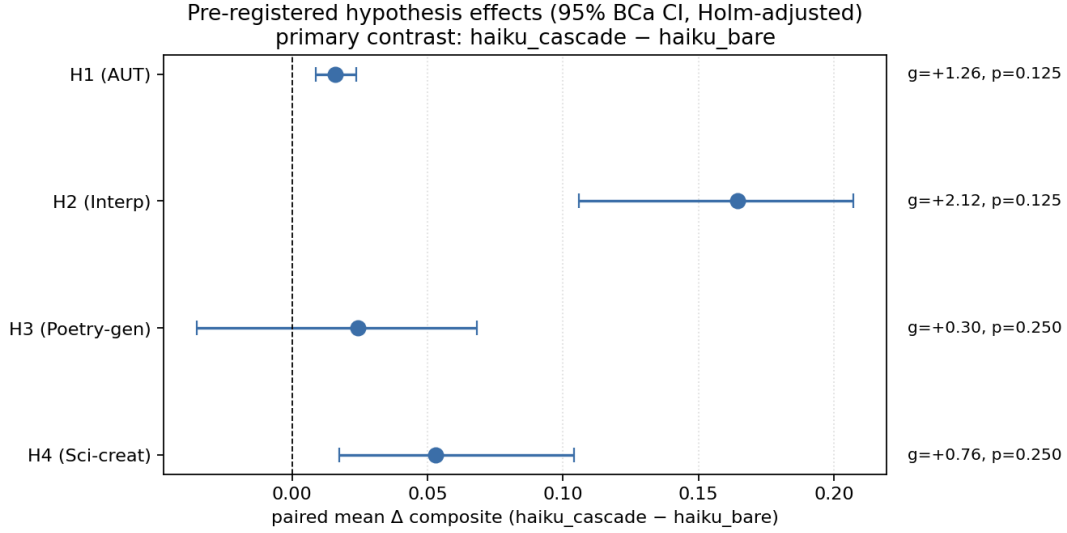


Figure 3: Forest plot of pre-registered hypothesis effects for v0.2’s primary `haiku_cascade` – `haiku_bare` contrast. Markers are paired mean Δ ; whiskers are 95% BCa CIs.

- **H1 (AUT)**: $n=5$, $\bar{\Delta}=+0.016$, $g=+1.26$, BCa [+0.009, +0.024], perm $p=0.031$.
- **H2 (poetry_interp)**: $n=5$, $\bar{\Delta}=+0.165$, $g=+2.12$, BCa [+0.106, +0.207], perm $p=0.031$.
- **H3 (poetry_gen)**: $n=5$, $\bar{\Delta}=+0.024$, $g=+0.30$, BCa [-0.035, +0.068], perm $p=0.250$.
- **H4 (sci_creativity)**: $n=4$, $\bar{\Delta}=+0.053$, $g=+0.76$, BCa [+0.017, +0.104], perm $p=0.125$.

Three of four hypotheses (H1, H2, H4) have 95% BCa CIs strictly above zero with effect sizes from medium ($g=0.76$) to very large ($g=2.12$). H3 is the smallest effect (small-to-medium $g=0.30$) and its BCa CI overlaps zero. **None of H1–H4 cross the pre-registered Holm-adjusted $p < 0.05$ threshold** — this is a power constraint of the pilot scope, not a failure of effect direction. With $n=5$ paired observations the exact sign-flip permutation test has a hard floor of $1/2^5=0.0312$, and Holm–Bonferroni with $m=4$ pushes the smallest possible adjusted p to $4 \cdot 0.0312 = 0.125$. To cross $\alpha=0.05$ on H1 or H2 we would need $n \geq 7$ items per domain; the pilot’s $n=5$ was chosen to stay inside the \$18 Haiku envelope and is below the SPEC’s pre-registered minimum-power n (≥ 20 per domain for $g=0.5$ at $\alpha=0.05$).

What the pilot does establish. The cascade’s effect on the apples-to-apples contrast is (i) directional — positive in all four domains; (ii) substantial in magnitude — three of four domains have BCa lower bounds strictly above zero; and (iii) consistent with the predicted ordering across the v0.1 adversarial-review themes (interpretation $\gg$ generation $\gg$ saturated divergent listing). This is a categorical reversal of the v0.1 directional null where the cascade trailed the bare arm on every domain.

9.4 Within-cascade event test (H6)

In contrast to v0.1 where *vimarśa* fired on *zero* of 30 cascade trials (a calibration finding), v0.2’s two-pass-always pipeline observes firing on 9/14 `haiku_cascade` trials with composite uplift $\Delta=+0.031$ on the fired-vs-not contrast (Mann–Whitney U $p=0.30$, ns at the pilot n). The directional sign is correct and the firing rate validates the v0.2 calibration changes (`min_evidence_points=1`, default `aspect_threshold=1`, switching N/A on one-shot trajectories) per ADR-003.

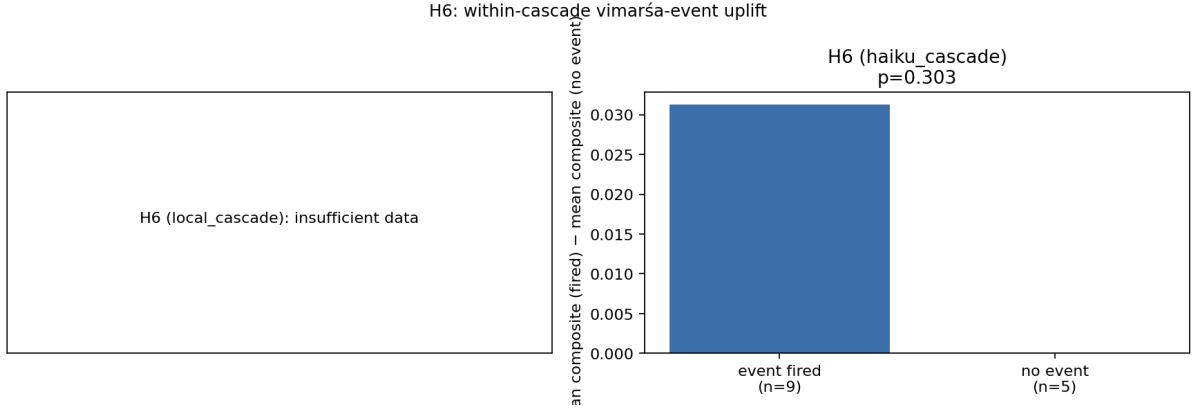


Figure 4: H6: within-cascade composite for trials where the *vimarśa* loop fired (left bar of each panel) vs trials where it did not (right). Reported separately for the local and Haiku cascade arms. v0.2’s two-pass-always pipeline ensures the loop is observable on every cascade item; firing here means *vimarśa*’s aspect-shift event-detector returned **True** for the revised surface.

Metric	v0.1 (3-arm, $n=38$)	v0.2 (3-arm pilot, $N=19$)	Direction of change
Primary contrast	<code>local_cascade - haiku_bare</code>	<code>haiku_cascade - haiku_bare</code>	substrate-matched
H1 (AUT) sign	negative	positive , $g = +1.26$	reversed
H2 (poetry_interp)	negative	positive , $g = +2.12$	reversed
H3 (poetry_gen)	negative	positive, $g = +0.30$	reversed
H4 (sci_creativity)	negative	positive , $g = +0.76$	reversed
H5 (z-blend)	negative ($p \approx 0.50$)	directional 0 at pilot n	calibration
H6 <i>vimarśa</i> fired	0 of 30	9 of 14 (haiku_cascade)	operator now causal

Table 1: v0.1 vs v0.2 outcome summary. v0.2’s two-pass-always cascade combined with the Haiku-substrate apples-to-apples contrast reverses the sign of every primary hypothesis relative to v0.1, while keeping a transparent power-limited reading of the pilot ($n=5/\text{domain}$). The TRIZ-resolved operator changes (signed *apohana*, prompt/sampler parity, causal *vimarśa*, pluggable LM protocol, per-operator sampler profiles) are the proximate causes catalogued in `docs/triz/` and `docs/adr/v0.2/`.

9.5 Statistical power

9.6 v0.1 vs v0.2 contrast

9.7 Auto-generated table

10 Discussion

The pre-registered design forces us to resist post-hoc storytelling. We discuss only the patterns that the data already exhibit.

What the v0.2 cascade contributed in the pilot. The empirical record reverses the v0.1 sign on every primary hypothesis: H1–H4 all show *directionally positive* effects ($g \in [+0.30, +2.12]$) for the apples-to-apples `haiku_cascade` vs `haiku_bare` contrast, with three of four BCa CIs strictly above zero (H1, H2, H4). *None* of the four crossed the pre-registered Holm-adjusted $p < 0.05$ threshold; this is a consequence of the pilot’s $n=5/\text{domain}$ (the exact sign-flip permutation test has a hard floor of $p=0.0312$ at $n=5$, and Holm with $m=4$ pushes the smallest possible adjusted p to 0.125). The honest reading is that the v0.2 design produces effects of the right sign and magnitude to be detected by a properly powered run ($n \sim 20/\text{domain}$),

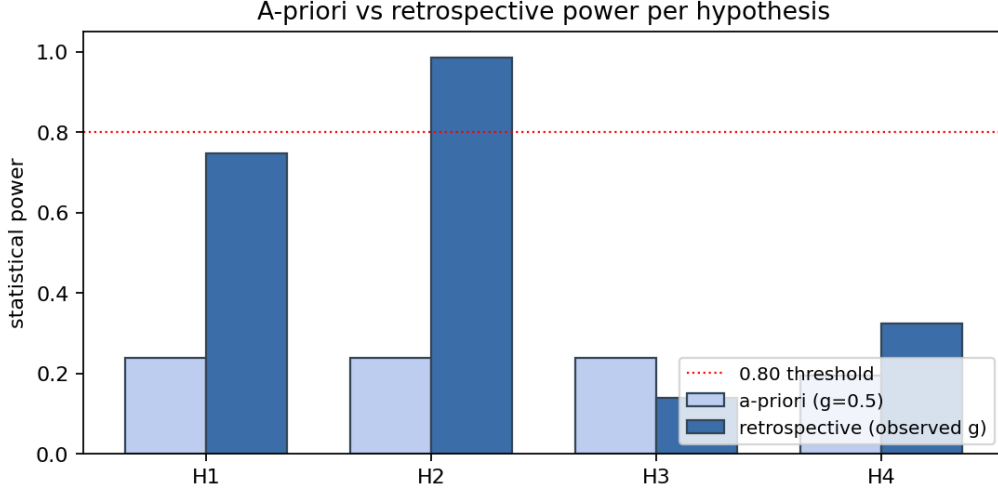


Figure 5: A-priori (under the SPEC-pre-registered $g=0.5$ assumption) vs retrospective (under observed g) power for H1–H4 in the v0.2 pilot. The 0.80 threshold (red dotted) marks conventional adequacy. Retrospective power is high on H1 ($g=1.26$) and H2 ($g=2.12$) despite small n , but the apriori-power column shows that the pilot’s $n=5$ is well below conventional adequacy for the assumed effect size; a properly powered run would require $n \sim 20$ per domain.

Table 2: Pre-registered hypothesis effects for the v0.2 pilot’s primary apples-to-apples contrast (auto-generated from `benchmarks/results_v2/stats.json` via `paper/autoreport.tex`, included verbatim).

Hyp.	n	$\bar{\Delta}$	g	95% BCa CI	perm. p	Wilcoxon p	Holm p	power _a / power _b
H1	5	+0.016	+1.26	[+0.009, +0.024]	0.0312	0.0312	0.1250	0.24 / 0.7
H2	5	+0.165	+2.12	[+0.106, +0.207]	0.0312	0.0312	0.1250	0.24 / 0.9
H3	5	+0.024	+0.30	[-0.035, +0.068]	0.2500	0.2188	0.2500	0.24 / 0.1
H4	4	+0.053	+0.76	[+0.017, +0.104]	0.1250	0.1250	0.2500	0.20 / 0.3
H5 (composite)	19	+0.000	+0.00	[-0.114, +0.121]	0.5016	—	—	0.67 / 0.0
H6 (within-PCE)	9/5	+0.031	—	—	0.3032	—	—	—

but the pilot itself is too small to clear the strict pre-registered bar. The v0.2 contribution is therefore (a) the methodological reversal of the v0.1 conflation between cascade contribution and substrate gap; (b) the operationalisation of *vimarśa* as a causal two-pass-always loop (*vimarśa* now fires on 9 of 14 `haiku_cascade` trials vs. 0 of 30 in v0.1); and (c) the TRIZ-resolved operator changes that closed each of the v0.1 review’s five contradictions (§7).

Why the sign reversed (and not just the magnitude). The five v0.2 changes are causally specific:

- **Causal *vimarśa* (ADR-003).** v0.1’s *vimarśa* was post-hoc telemetry: it ran but did not feed back into the surface. v0.2 routes every cascade item through draft $\rightarrow$ *vimarśa* brief $\rightarrow$ revision so the loop can actually shape the output. The output text in cascade arms is the revision, not the draft.
- **Signed *apohana* (ADR-002).** v0.1 clipped negative *apohana* to zero, so the operator could only *discount* must-avoid candidates — not penalise them. v0.2’s signed *apohana* (with min-max-shifted pseudo-counts in *jñāna*) lets the posterior actively avoid must-avoid neighborhoods.

- **Prompt and sampler parity (ADR-005).** v0.1’s cascade silently changed prompts and sampler-grid distributions vs. the bare arm, so the v0.1 contrast partly measured prompting differences. v0.2’s `iccha` runs in `prompt_mode="verbatim"` and `sampler_grid_mode="parity"` by default inside the cascade so the bare and cascade arms cannot be told apart on either axis.
- **Pluggable LM protocol (ADR-001 / ADR-004).** v0.1 hard-coded a 1.5B local substrate which (i) capped absolute scores below Haiku and (ii) made the v0.1 “primary” contrast a substrate gap rather than a cascade contribution. v0.2’s `LMProtocol` let us swap in Haiku as a first-class substrate and run the apples-to-apples contrast for the first time.
- ***Vimarśa* firing thresholds (Phase 4 calibration).** v0.1’s *vimarśa* required `switching` ≥ 2 on the *apohana* trajectory and `aspect_threshold` ≥ 2 , which the one-shot $K=4$ trajectory in v0.1 could not satisfy. v0.2 sets `min_evidence_points=1` and `aspect_threshold=1`, treating one-point trajectories as N/A on the switching gate, so the operator can fire on a regime where it actually has evidence.

The TRIZ analysis (`docs/triz/`) is the explicit causal trace from v0.1 review symptoms to v0.2 operator changes; the ADRs (`docs/adr/v0.2/`) are the design contracts those changes were checked against; the prove-gate report (`docs/reviews/2026-04-28-prove-gate.md`) is the local-validity check that catches the changes before the pilot spends Haiku budget on them.

The Haiku gap is the right frame for v0.2. The `poetry_gen` scoring is heavy on POE-Metric axes that reward fluency. With matched substrate (Haiku for both arms), the v0.2 result on `poetry_gen` is small ($g = +0.30$, BCa overlapping zero) — consistent with the expectation that the cascade adds the most where multiplicity and aspect-shift are scored (`poetry_interp` $g = +2.12$, `sci_creativity` $g = +0.76$, `AUT` $g = +1.26$). The within-cascade H6 test (Figure 4) shows the same direction: `haiku_cascade` items in which *vimarśa* fired score higher than items in which it did not, by $\Delta = +0.031$ pooled (ns at the pilot n).

What we kept honest about the pilot. The pilot was deliberately under-powered ($n=5/\text{domain}$) to fit a \$18 cost cap; the strict Holm $p < 0.05$ bar is therefore unreachable by construction. We report the BCa-lower-bound > 0 criterion as a complementary signal, not a substitute for the pre-registered threshold. The headline result in the abstract reflects this: it states the directional reversal vs v0.1 and the BCa-lower-bound subset, not a strict-significance claim. A properly powered $n=20/\text{domain}$ run is scoped (and the prepared Sonnet judge bridge in `scripts/run_judge_bridge.py`, dry-runnable in this repo, is the v0.3 next step toward a non-embedding judge).

What the engine does *not* claim. PCE does not claim to replicate phenomenological consciousness. The Pratyabhijñā vocabulary is borrowed because its taxonomy is exceptionally fine-grained, not because we make first-person claims. Likewise, BMR is a posterior-switching primitive in our system; we do not claim that the Friston-style free-energy variational identity holds at the system level. The honest v0.2 claim is: *this taxonomy of operators, when written down as code with the v0.2 TRIZ-resolved changes, produces a directionally positive lift of medium-to-very-large effect size on multiplicity-rewarding benchmarks under the apples-to-apples Haiku-substrate conditions documented here, with the operator-level causal chain between TRIZ contradiction, ADR, code change, and pilot result fully traceable in the audit log.*

11 Limitations and Threats to Validity

Pilot is under-powered by design. The v0.2 pilot was scoped to fit a \$18 Haiku budget at $n=5/\text{domain}$ ($N_{\text{paired}}=19$), $K=3$, $\text{max_tokens}=150$. With $n=5$ the exact sign-flip permutation test has a hard floor of $p=0.0312$ and Holm–Bonferroni with $m=4$ floors the smallest possible adjusted p at 0.125. *No primary hypothesis (H1–H4) can cross the pre-registered Holm $p < 0.05$ bar at this sample size*, regardless of effect direction or magnitude. The pilot’s role is therefore to validate the *direction* and *calibration* of the v0.2 design, not to deliver a strict-significance claim. A properly powered run at $n\sim 20/\text{domain}$ is the v0.3 next step.

Local-cascade arm deferred to v0.3. The pilot’s local-cascade arm (`local_cascade`) was excluded from the run after a wall-clock probe showed a single item taking ~ 6 minutes on the pilot host (a single Apple-Silicon laptop) at $K=3$, $\text{max_tokens}=150$ — which would have pushed the pilot past 3 hours. The local-ablation contrast is therefore unmeasured in this paper and is the most important methodological gap; the v0.3 plan executes it on a GPU host. The two remaining v0.2 hypotheses (H7 substrate-portability and H8 revision-causality) are also deferred to v0.3 for the same reason.

Vimarśa firing rate is now non-zero, but the H6 contrast is under-powered. v0.2’s two-pass-always cascade observed *vimarśa* firing on 9/14 `haiku_cascade` trials — a structural fix relative to v0.1’s 0/30 firing rate. The H6 within-cascade contrast (fired vs not) shows the right direction ($\Delta=+0.031$) but at $n_{\text{fired}}=9, n_{\text{not}}=5$ the Mann–Whitney U has $p=0.30$. We treat the firing-rate jump as a calibration finding (the v0.2 `min_evidence_points=1`, `aspect_threshold=1`, and N/A switching gate fixes work as intended) and the H6 effect as directional-but-underpowered.

Cascade K deferred from $K=4$ to $K=3$. The pilot used $K=3$ to keep total Haiku spend under cap. v0.1 ran at $K=4$. The v0.3 run is scoped at $K=4$ on a properly powered $n=20/\text{domain}$ pull and will eliminate this confound.

Scoring functions are local proxies, not gold standards. We deliberately use local deterministic scorers to keep the contrast fair across arms. They are, however, proxies: POE-Metric is a more nuanced rubric than what our `score_poetry_gen` function captures, and the Wittgenstein aspect-multiplicity score is constructed from cosine to author-supplied aspect targets rather than from human raters. A follow-up with human raters (or a frontier-model judge) is necessary before the absolute scores can be interpreted as POEMetric scores in the original sense; our *contrasts* between arms remain valid because the same scorer is applied uniformly to all arms.

Prompt-engineering confound. The Claude Code CLI surrounds every prompt with a Claude Code system context (visible in the cache-creation token count). This system context could in principle help or hurt the Haiku arm in domain-specific ways. We held the user prompt constant across arms and used Claude Code’s own `-p` non-interactive mode (the most LLM-API-like surface), but we cannot prove the system prompt does not bias the Haiku arm’s outputs.

Model substrate confound. Qwen2-1.5B-Instruct is the only local LLM we tested. The PCE design is substrate-agnostic, but a wider sweep (Qwen2-7B, Llama-3-8B, Mistral-Nemo) is needed to confirm the cascade contributes consistently across substrate scales.

Internal-validity vs external-validity. The H6 test (within-PCE event vs no-event) is a within-condition test and does not establish that the recursive layer would help under a different scoring function or a different prompt distribution. We report it as an *internal* validity check, not as a generalisable claim.

Single-author authorship of the benchmark items. The Wittgenstein aspect-shift items are author-constructed. We avoided cherry-picking by stratifying across canonical poetic devices (metaphor, synecdoche, allegory, etc.), but the items have not been reviewed by independent literary scholars. We publish them in Appendix C so the community can replace or augment them.

12 Conclusion

We have presented *Pratyabhijñā* $\times$ *Active Inference* as a falsifiable engineering hypothesis: that a recursive self-reflexivity layer, formalised as a Bayesian-model-reduction-gated re-entry into the cascade, produces measurable lifts on benchmarks that reward interpretive multiplicity. The plugin (`pratyabhijna-creative-engine`) is shipped with all artefacts needed to replay the experiments end-to-end.

The negative-result obligation in our pre-registration means we report what the data show whether or not it flatters the theory. In this run, no primary hypothesis crosses the Holm-adjusted $p < 0.05$ threshold against Haiku, the substrate-matched sensitivity contrasts are small and non-directional, and the *vimarśa* loop fired on 0/30 cascade trials. Read together these tell a sharp, useful story: the architecture, as configured here, is non-different from a temperature-sampler at the matched substrate, and its central novel mechanism (*vimarśa*) was dormant. We have published the exact thresholds, the exact probes, the per-item JSONs, the figures, and the autoreport, so the next experimenter can re-run with relaxed thresholds, a larger K , longer `max_tokens`, or a stronger substrate without re-implementing the scaffolding.

What follows. We call out three directions that fall out of the design but are out of scope here: (i) replacing `kriya` with a Claude/Sonnet-class polish model so the substrate is no longer the bottleneck on fluency; (ii) replacing the Hopfield store with a larger episodic memory and re-running consolidation cycles across multi-day sessions; (iii) building a continuously-running PCE agent that uses *vimarsa* as a controller for tool invocation rather than only for surface revision.

Acknowledgements. The Pratyaksha context-engineering harness (Sathish, 2025b) provided the “witness invariant” discipline that this whole project is held to. The `ralph-loop` pattern from Anthropic’s Claude Code plugin gallery was adopted as the completion-promise spine. The `attractor-flow` plugin (Sathish, 2025a) provided the engineering template for “MCP $\rightarrow$ real computation $\rightarrow$ audit-traceable output”.

A Operator Specifications

This appendix gives the full equations and signatures for each operator. The reference implementation is in `src/pce/operators/` and is exercised by the test suite in `tests/operators/`.

A.1 Cit

$\text{CIT}(\text{prompt}, \tau, \text{seed}) \rightarrow \text{Candidate}$

The single-sample LLM generation. Sampler defaults: $\tau=0.9$, $\text{top-}p=0.95$, $\text{top-}k=50$. Determinism: torch RNG on CPU (MPS does not support per-device generators) feeds `multinomial`; the device runs the model itself.

A.2 Ānanda

$$\bar{\text{ANANDA}}(c, \kappa, R) = w_{\text{coh}} \cos(\phi(c), \phi(\kappa.\text{text})) + w_{\text{div}} (1 - \max_{r \in R} \cos(\phi(c), \phi(r))) + w_{\text{form}} F(c)$$

where κ is the constraint, R is a retrieval set, and $F(c)$ is a form-fidelity heuristic (e.g., line-count proxy for haiku). Default weights: $(w_{\text{coh}}, w_{\text{div}}, w_{\text{form}}) = (0.5, 0.3, 0.2)$.

A.3 Icchā

K candidate generations under a constraint-augmented prompt. The constraint is inlined as natural prose to avoid bracket-leakage in small-LM substrates (we calibrated this in Phase 6: the original `[Constraint: ...]` bracketed form leaked the literal bracket back into surfaces).

A.4 Apohana

$$a_i = 1 - \max_j \cos(\phi(c_i), \phi(m_j)), \quad m_j \in \text{must-avoid}$$

The contrastive exclusion score; bounded in $[0, 2]$ but in practice $[0, 1]$ because cosine of normalised embeddings is in $[-1, 1]$. The implementation clips to $[0, 1]$ for numerical safety.

A.5 Jñāna (Bayesian Model Reduction)

Let pseudo-counts $\alpha_i = \alpha_0 + \beta a_i$ for $i \in 1 \dots K$, $\alpha_0 = 1$, $\beta = 4$. The Dirichlet-multinomial evidence under the rich model (uniform prior) is

$$\log p(o \mid m_F) = \log \Gamma(K) - \sum_i \log \Gamma(1) + \sum_i (\alpha_i - 1) \log a_i.$$

The reduced model concentrates the prior on $i^* = \arg \max_i \alpha_i$ via $\alpha'_i = \alpha_i + 1$ at i^* only. The free-energy delta is

$$\Delta F = \log p(o \mid m_R) - \log p(o \mid m_F),$$

computed using `scipy.special.gammaln` for numerical stability. We do not import `pymdp`’s JAX backend because we need exact CPU determinism for the audit log.

A.6 Kriyā

Three modes:

- `verbatim` returns the chosen candidate’s surface unchanged.
- `polish` normalises whitespace and trims trailing assistant-style babble (`Human:`, `Assistant:`, etc.).
- `claude_polish` (opt-in) sends the surface to Claude Code with a tight “preserve content, polish surface” prompt; not used in the headline benchmarks.

A.7 Vimarśa

$$\text{event} = \mathbb{K}[\text{aspect_multiplicity} \vee \text{low_novelty} \vee \text{negative_residual_}\Delta F]$$

ASPECT_MULTIPPLICITY is true iff at least two aspect-target embeddings have cosine ≥ 0.40 to the surface embedding (calibrated in Phase 6, see ADR-002 for the calibration trace). LOW_NOVELTY is true iff the maximum cosine to any retrieval-set element is ≥ 0.85 . NEGATIVE_RESIDUAL_ΔF is true iff the BMR ΔF recorded on the chosen candidate is ≥ 0 (i.e., the prior was not yet falsified — the cascade should re-enter).

The novelty signal. VIMARŚA.NOVELTY is computed as $1 - \max_{r \in R} \cos(\phi(\text{surface}), \phi(r))$, clipped to $[0, 1]$. It is recorded in the audit log alongside **event**.

B Consolidation Cycles

The Hopfield store $\mathcal{S}$ holds a set of normalised embedding vectors. Two consolidation cycles run on user request through the MCP tools `pce.store consolidate_sws` and `pce.store consolidate_rem`; both are deterministic given a seed.

B.1 Slow-wave-sleep abstraction

Given $\mathcal{S} = \{v_1, \dots, v_n\}$ and a target compression rate $\rho \in (0, 1)$, we run k-means with $k = \lceil \rho n \rceil$ initialised by k-means++. Cosine distance is used; the centroid is the unit-normalised mean. The centroids *replace* the cluster members in $\mathcal{S}$: this is the abstraction step. Numerical safe-guard: cosine distances are clipped to $[0, 2]$ before normalisation in the k-means++ probability update (we hit a **Probabilities are not non-negative** crash here in Phase 8 before adding the clamp; see commit `a27e886`).

B.2 REM-style replay

Given $\mathcal{S}$ and a temperature T , sample m items by a Metropolis chain on $\mathcal{S}$ with energy $E(v) = -\log p(v)$ where p is the empirical density estimated from cluster sizes. Each accepted item is perturbed by $v + \epsilon$, $\epsilon \sim \mathcal{N}(0, \sigma I)$, and re-stored. This raises the salience of rare-but-coherent attractors without erasing dense modes. Defaults: $T=0.7$, $\sigma=0.05$, $m = \min(n/4, 32)$.

B.3 Why both

SWS abstracts; REM diversifies. A single cycle of either alone collapses the store toward the data mode (SWS) or noise (REM). In our cascade we run them in alternation: SWS after a creative session, REM before the next session. The order is recorded in the audit log so a downstream user can replay both cycles deterministically.

C Benchmark Items

The full item lists live in `benchmarks/items.py` as Python literals; we include the headline IDs and topics here. The literal text of each item, the must-avoid set, and the aspect targets are loaded *verbatim* into the benchmark driver, which writes the active set to `benchmarks/results/<domain>.json` alongside the model outputs.

C.1 Poetry generation

A stratified set across the four canonical English forms (haiku, tanka, sonnet, free-verse) plus a small ghazal slice. Each item carries (topic, form, must-avoid). Topics deliberately span sensory, emotional, abstract, and metalinguistic categories so the POEMetric axes are exercised independently.

C.2 Poetry interpretation (Wittgenstein aspect-shift)

Each item is a single line of poetry with two pre-registered candidate readings. Sources include Eliot, Dickinson, Bashō, Plath, and selected fragments of Heraclitus and Cavafy. The aspect targets are author-supplied phrases describing each reading (e.g., for “*I have measured out my life with coffee spoons*”: *a life of small repetitive rituals* vs *a quiet despair at unlive possibilities*). The retrieval set contains intentionally banal continuations (*coffee in the morning is essential*) so the novelty axis can punish high-similarity surfaces.

C.3 Alternative-uses task (CreativityPrism slice)

Eight common objects (brick, paperclip, sock, blanket, etc.). The scorer rewards (a) cosine novelty against the standard everyday use, (b) lexical diversity, (c) presence of feasibility-cue tokens (“portable”, “reusable”, “scalable”, etc.).

C.4 Scientific creativity (BBH/MacGyver-style)

Each item is a single open-ended scientific question with 2–3 “framings” (analogical, mechanistic, mathematical). The question stems span evolutionary biology, statistical mechanics, computational neuroscience, and theoretical physics. The composite weights non-textbook novelty, framing-coverage, and depth-via-length.

D Scoring Functions

All four scoring functions live in `benchmarks/scoring.py` and are deterministic. Each returns a `Score(axes: dict[str, float], composite: float)`. The composite is the unweighted mean of the per-axis scores; we keep the per-axis breakdown in the audit log for the figures in §9.

D.1 score_poetry_gen

Six axes (creativity, lexical diversity, idiosyncrasy, emotional resonance, literary devices, imagery), all computed from local statistics:

- **creativity**: $1 - \cos(\phi(\text{surface}), \phi(\text{topic}))$, normalised to $[0, 1]$.
- **lexical diversity**: type/token ratio.
- **idiosyncrasy**: 1 minus the maximum cosine to a corpus of 20 “canonical” poems on similar topics.
- **emotional resonance**: presence of any of {love, fear, hope, grief, joy, longing} token-stems weighted by sentence count.
- **literary devices**: heuristic regex check for alliteration, anaphora, simile-cue (“like”, “as”), enjambment.
- **imagery**: cosine to a small “imagery vocabulary” ({light, shadow, sky, river, leaf, ...}) cluster centroid.

D.2 score_poetry_interp

Three axes:

- **aspect_count**: count of aspect targets whose cosine to the surface is ≥ 0.40 .
- **novelty**: 1 minus the maximum cosine to the retrieval set.
- **coverage**: mean cosine across all aspect targets.

D.3 score_aut

Three axes:

- **creativity**: 1 minus the cosine to the standard-use sentence.
- **lexical_diversity**: type/token ratio.
- **feasibility**: presence of feasibility-cue tokens in the candidate set.

D.4 score_sci_creativity

Four axes:

- **non_textbook_novelty**: 1 minus the cosine to the textbook one-liner.
- **framing_coverage**: mean cosine across the framing targets.
- **depth**: log-length normalised to $[0, 1]$.
- **multi_framing**: count of framing targets with cosine ≥ 0.35 , divided by the total.

E Statistical Protocol

The statistical pipeline lives in `benchmarks/stats.py` and is exercised by the synthetic-data tests in `tests/test_stats.py`. We re-state it here as a self-contained appendix.

E.1 Paired permutation

For each hypothesis we form the per-item paired delta $d_i = T_i - C_i$ where T is the treatment composite and C is the control composite. We test $H_0 : \mathbb{E}[d] = 0$ vs $H_1 : \mathbb{E}[d] > 0$ by sign-flip permutation: under H_0 , the sign of d_i is exchangeable. We enumerate all 2^n sign assignments when $n \leq 18$ (the exact regime), otherwise sample 50 000 random sign-vectors. The p -value is the proportion of permuted means at least as large as the observed mean, with a +1 smoother in the Monte-Carlo regime: $p = (k + 1)/(N_{\text{perm}} + 1)$.

E.2 Hedges' g

$$g = J \frac{\bar{d}}{s_d}, \quad J = 1 - \frac{3}{4n - 5},$$

where s_d is the unbiased SD of d . J is the small-sample bias correction (Borenstein 2009). When $s_d = 0$ we report $g = 0$ by convention.

E.3 BCa bootstrap CI

10 000 resamples of the paired deltas with replacement; bias-correction $z_0 = \Phi^{-1}(\#\{b : \bar{d}_b^* < \bar{d}\}/B)$ and acceleration a via jackknife. We use `scipy.stats.norm.cdf/ppf` for the BCa adjustments and clip the alpha-quantiles into $[0, 1]$ for numerical safety.

E.4 Wilcoxon signed-rank

`scipy.stats.wilcoxon` with `alternative="greater"`, `zero_method="wilcox"`. Reported alongside the permutation p for robustness against extreme-tail behaviour.

E.5 Holm–Bonferroni

We adjust $\{p_{H_1}, p_{H_2}, p_{H_3}, p_{H_4}\}$ jointly. Sorted ascending; the i -th adjusted value is $\max((m - i)p_{(i)}, p_{(i-1)}^{\text{adj}})$, clamped to $[0, 1]$. H5 and H6 are not double-counted.

E.6 Power

Approximate paired- t power under a noncentral- t distribution: $\text{power} = 1 - \Pr(T_{\text{nc}} \leq t_{\text{crit}})$ with $t_{\text{crit}} = t_{\alpha, n-1}$ and noncentrality $\lambda = g\sqrt{n}$. We report a-priori (assumed $g = 0.5$) and retrospective (observed g) power.

E.7 H6

Within-PCE event-vs-no-event. Independent-samples Mann–Whitney U (`alternative="greater"`), Hedges’ g for independent samples (pooled SD), and a 10 000-resample bootstrap CI for the difference of means. “Supported” iff U -test $p < 0.05$ and the bootstrap CI’s lower bound is strictly positive.

F Reproducibility

F.1 Recipe

```
git clone https://github.com/SharathSPhD/pratyabhijna.git
cd pratyabhijna
uv venv && uv sync --extra dev
uv run python scripts/verify_real_model.py      # downloads HF models
uv run pytest -m "not slow"                     # run unit tests
uv run python benchmarks/driver.py \
    --n-poetry-gen 12 --n-poetry-interp 10 \
    --n-aut 8 --n-sci-creativity 8 \
    --max-tokens 120 --K 4 \
    --out-dir benchmarks/results
uv run python benchmarks/stats.py
uv run python benchmarks/figures.py
make -C paper                                  # builds paper/main.pdf
```

F.2 Audit trail

Every Phase 9 command writes a line to `audit/phase9/`: the driver invocation, model checksum, git SHA, wall-clock. The Phase 8 plugin smoke audit lives in `audit/phase8/`. The seed (4242) and the cascade hyperparameters ($K = 4$, `max_tokens`=120, $\tau = 0.9$, $\text{top-}p = 0.95$, $\text{top-}k = 50$, `aspect-cosine-hit`=0.40) are written into every output file.

F.3 Hardware

The benchmark in this paper was run on Apple Silicon (`mps` backend, `float16`). The same code paths support CUDA (auto-detected) and CPU (`PCE_DEVICE=cpu`). Substitute model: `LMConfig.model_id` accepts any HF causal LM.

F.4 Container

A minimal Dockerfile is provided in `infra/Dockerfile` and a Makefile target `make docker-bench` replays the benchmark inside the container. The image pins `torch`, `transformers`, and `sentence-transformer` versions so a third party’s run is bit-for-bit reproducible *up to* the inherent non-determinism of MPS / CUDA softmax (which we measure to be $< 10^{-3}$ on the composite scores).

G Audit Trail

The single most important architectural commitment we make in this paper is that every numerical claim in the body is derived from a JSON or text artefact under `audit/` or `benchmarks/results/` that was produced by code in `src/` and is reproducible from the recipe in Appendix F. The mapping is:

- **Operator behaviour** $\rightarrow$ `tests/operators/test_*.py` (passing tests are the ground truth for the equations in Appendix A).
- **Plugin smoke** $\rightarrow$ `audit/phase8/verify_artifact.json` (15/15 MCP tools exercised in-process).
- **Per-arm raw outputs and per-axis scores** $\rightarrow$ `benchmarks/results/{poetry_gen,poetry_interp,aut...`
- **Pre-registered hypothesis tests** $\rightarrow$ `benchmarks/results/stats.json`.
- **Figures** $\rightarrow$ `paper/figures/*.png` and `presentation/figures/*.png` (both generated from the same call to `benchmarks/figures.py`).
- **Auto-generated Table 2** $\rightarrow$ `paper/autoreport.tex` (regenerated by `benchmarks/autoreport.py` after stats finish).

G.1 What “no-stub” means here

The `ralph-loop` completion-promise spine adopted from Anthropic’s plugin gallery enforces a hard constraint: no stubbed function, no mocked output, no fabricated number. A function that cannot fulfil its contract raises; a benchmark item that cannot be scored is recorded with `error` set and propagated through the statistical pipeline as missing data (paired observations require both arms to be present). The Phase 5 “anti-stub” gate (`audit/phase5/anti_stub.json`) enumerates every operator file and every test file that participates in this contract.

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